

Cars and the IoT

Cars and other vehicles are perhaps the most obvious of 'things' climbing onto the IoT bandwagon today

Autonomous self-driving cars might be a decade away from mainstream use, but almost every automaker is now focusing on smart connected cars, and these are proving to be quite awesome and useful as well. Gartner Analyst Thilo Koslowski describes connected vehicles as cars that access, consume, create, enrich, direct and share digital information between businesses, people, organisations, infrastructures and things. "Those 'things' include other vehicles, which is where the Internet of Things becomes the Internet of Cars," he writes in one of his articles.

Industry experts believe that the elite have already started looking at cars as mobile devices, and began evaluating models by the extent of connectivity, car management, security and social apps, and so on. Very soon, all users will begin using these as critical aspects of evaluation. So, what can connected cars do for you, for others, for the environment?

1. Cars will have better and smarter navigation systems compared to today. They will be able to assess alternative routes, 'talk' to other cars and find out how good or bad a route is, and so on, to arrive at the best route for you.

2. They will be capable of handling voice-response commands. They will also let you connect with Pandora, Stitcher and other apps on your phone via the car's voice-recognition interface.

3. Cars will connect with wearable sensors to monitor your health condition. In case of trouble, the car would park on the side and buzz your healthcare specialist. Ford, for example, is working on a car with sensors and an assisted driving mode that makes it safer for people with seizure to drive.

4. They will help you find parking lots!

5. Connected cars will know more about their users, and change modes,

suggest routes, etc., based on users' preferences.

6. Car sharing could be automated, rather than use a traditional ask-and-approve system. Sensors in cars, tied to GPS positioning, could let a backend network know where a car is parked at any given moment, and plan car-pooling based on that information.

7. Cars will help the environment by moderating air-conditioner use, aiding efficient driving with fuel-save modes, reducing traffic, and so on.

8. Car-to-car communication and navigation together with other in-car safety features would help prevent accidents altogether.

Tesla, Ford, Toyota, Honda, almost all the world leaders in cars are working on smart cars, and autonomous ones as well. Silicon specialists like Intel, NXP and Texas Instruments are also entering the automotive value chain. Soon, telecom players will be providing specialised 'plans' for auto-users just as they did for the BlackBerry or iPhone.

More than anything else, consumers are also showing a strong interest in the features of a connected vehicle. Recent Gartner researches focused on US auto consumers show that 46 per cent are interested in safely accessing mobile applications like on-demand wireless maps, software updates, parking search, local searches, etc. inside the vehicle. Nearly 40 per cent would also opt for remote diagnostic capabilities that alert them when parts need replacement. More than one-third are interested in a self-driving, autonomous vehicle, and thirty per cent are likely to opt for a vehicle that allows them to tether their smartphone to get Internet connection there.

So, here we are, at the cusp of the Internet of cars!

Counting the eggs in your fridge



GE has invented a sub-\$15 IoT device that lets you remotely check how many eggs you have in your fridge and decide whether to buy more. The Egg Minder is a very simple device. It is a tray with 14 egg cups, each with a sensor that can check whether it contains an egg. This information is transmitted wirelessly to a server in the cloud, which can then feed the data to an app on your smartphone.

Solve a problem



To be successful, an IoT product must be more than a passing fad; it should be a utilitarian product that solves a real problem for people. Jason Johnson, leader of the IoT consortium, said in a recent media interview, "For the Internet of Things to work, there must be a Plan B when it doesn't work." The August Smart Lock is a simple example that helps us understand this better. It is a smart door lock that recognises who is approaching your home and lets you open the door on command. You can give permanent or temporary access to some people and even work around regular access rules in times of emergencies. More than anything, it has a Plan B: it lets you open the lock with an old-fashioned key if the power goes out!

When simple things fail...

While there is much talk about the IoT, there are times when simple communication failures make you wonder where we stand in relation to such a large dream. In a recent TED talk on how future jobs will look like, famous economist Andrew McAfee commented: "There's no shortage of dystopian visions about what happens when our machines become self-aware and they decide to rise up and coordinate attacks against us... I'm going to start worrying about the day my computer becomes aware of my printer."

Read more about the Internet of Things at electronicsforthings.com